3 (a) Gravitational force provides for centripetal force.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$ M1

$$\frac{GM}{r} = v^2$$

$$v = \sqrt{\frac{GM}{r}}$$ A0

(b)(i) $$v = \frac{2\pi r}{T} \Rightarrow T = \frac{2\pi r}{v}$$ C1

$$\frac{T_A}{T_B} = \frac{r_A}{r_B} \times \frac{v_B}{v_A}$$

$$= \frac{r_A}{r_B} \times \sqrt{\frac{r_A}{r_B}} \qquad \text{since } v = \sqrt{\frac{GM}{r}} \propto \sqrt{\frac{1}{r}}$$ C1

$$= \frac{1.3 \times 10^8}{2.2 \times 10^{10}} \times \sqrt{\frac{1.3 \times 10^8}{2.2 \times 10^{10}}}$$

$$= 4.5 \times 10^{-4}$$ A1

(ii)

$$\omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega}$$

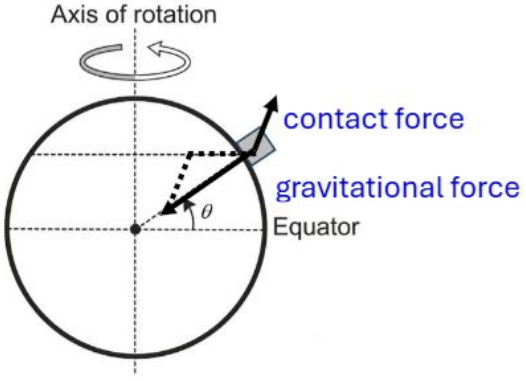
$$T_A = \frac{2\pi}{1.7 \times 10^{-4}} = 3.7 \times 10^{-4}$$ C1

$$\frac{T_A}{T_B} = 4.5 \times 10^{-4}$$

$$T_B = \frac{T_A}{4.5 \times 10^{-4}}$$

$$= \frac{3.7 \times 10^{-4}}{4.5 \times 10^{-4}}$$

$$= 8.2 \times 10^7 \text{ s}$$ A1



• Gravitational force pointing towards the centre of planet B1

• Contact force by ground on block. (Accept if normal contact force and friction force are drawn separately). B1

Resultant of the 2 forces must act towards centre of circle